

# Encrypted Network Traffic Forensic Report

**CONFIDENTIAL**

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|---|---|
| Case ID | NS-2026-001 |
| Analyst | Security Analyst |
| Date Generated | 2026-07-16 13:36:57 UTC |
| Analysis Mode | Forensic (PCAP) |
| Source | knoten_in_the_wire.pcap |
| Total Packets Analyzed | 5,506 |
| Total Flows | 5,423 |
| Anomalies Detected | 272 |
| Tool Version | NetSentinel v1.0 |

# 1. Executive Summary

**Overall Risk Level:** MEDIUM

NetSentinel analyzed 5,506 packets across 5,423 network flows. The Isolation Forest algorithm flagged 272 anomalous flows (5.0% of traffic). See detailed findings below.

**Key Findings:**

- 272 anomalous flows detected out of 5,423 total flows
- 0 JA3 fingerprints matched against known malware signatures
- 0 IP addresses flagged for suspicious reputation
- Analysis duration: N/A

# 2. Threat Timeline

Chronological sequence of detected anomalous events during the analysis window.

| # | Timestamp | Source | Destination | Protocol | Risk Score | Trigger |
|---|---|---|---|---|---|---|
| 1 | 2032-12-08 12:17:39 | 192.168.33.10:80 | 192.168.33.30:59637 | TCP | 93.4 | Isolation Forest |
| 2 | 2032-12-08 12:18:34 | 192.168.33.1:53 | 192.168.33.30:61066 | DNS | 63.3 | Isolation Forest |
| 3 | 2032-12-08 12:17:26 | 192.168.33.10:80 | 192.168.33.20:63745 | TCP | 90.5 | Isolation Forest |
| 4 | 2032-12-08 12:17:20 | 192.168.33.10:80 | 192.168.33.30:50306 | TCP | 53.7 | Isolation Forest |
| 5 | 2032-12-08 12:18:01 | 192.168.33.1:53 | 192.168.33.40:57788 | DNS | 85.1 | Isolation Forest |
| 6 | 2032-12-08 12:18:49 | 192.168.33.1:53 | 192.168.33.20:51067 | DNS | 53.0 | Isolation Forest |
| 7 | 2032-12-08 12:18:12 | 192.168.33.1:53 | 192.168.33.60:59557 | DNS | 77.6 | Isolation Forest |
| 8 | 2032-12-08 12:18:18 | 192.168.33.1:53 | 192.168.33.30:59114 | DNS | 63.3 | Isolation Forest |
| 9 | 2032-12-08 12:17:20 | 192.168.33.10:80 | 192.168.33.40:51358 | TCP | 84.9 | Isolation Forest |
| 10 | 2032-12-08 12:18:42 | 192.168.33.1:53 | 192.168.33.30:49483 | DNS | 63.3 | Isolation Forest |
| 11 | 2032-12-08 12:17:15 | 192.168.33.10:80 | 192.168.33.50:50527 | TCP | 90.8 | Isolation Forest |
| 12 | 2032-12-08 12:18:48 | 192.168.33.1:53 | 192.168.33.60:59763 | DNS | 51.3 | Isolation Forest |
| 13 | 2032-12-08 12:18:59 | 192.168.33.1:53 | 192.168.33.30:51273 | DNS | 60.3 | Isolation Forest |
| 14 | 2032-12-08 12:16:52 | 192.168.33.10:80 | 192.168.33.60:57544 | TCP | 89.3 | Isolation Forest |
| 15 | 2032-12-08 12:17:13 | 192.168.33.10:80 | 192.168.33.30:56238 | TCP | 53.7 | Isolation Forest |
| 16 | 2032-12-08 12:19:12 | 192.168.33.10:80 | 192.168.33.40:52606 | TCP | 91.9 | Isolation Forest |
| 17 | 2032-12-08 12:18:02 | 192.168.33.1:53 | 192.168.33.30:57026 | DNS | 60.3 | Isolation Forest |
| 18 | 2032-12-08 12:19:10 | 192.168.33.10:80 | 192.168.33.20:61314 | TCP | 89.6 | Isolation Forest |
| 19 | 2032-12-08 12:18:14 | 192.168.33.1:53 | 192.168.33.20:60827 | DNS | 53.0 | Isolation Forest |
| 20 | 2032-12-08 12:17:24 | 192.168.33.10:80 | 192.168.33.30:49978 | TCP | 53.7 | Isolation Forest |
| 21 | 2032-12-08 12:18:37 | 192.168.33.1:53 | 192.168.33.20:54926 | DNS | 53.0 | Isolation Forest |
| 22 | 2032-12-08 12:19:03 | 192.168.33.1:53 | 192.168.33.30:54101 | DNS | 52.4 | Isolation Forest |
| 23 | 2032-12-08 12:17:36 | 192.168.33.10:80 | 192.168.33.20:52686 | TCP | 93.9 | Isolation Forest |
| 24 | 2032-12-08 12:16:48 | 192.168.33.10:80 | 192.168.33.60:51384 | TCP | 91.2 | Isolation Forest |
| 25 | 2032-12-08 12:18:51 | 192.168.33.1:53 | 192.168.33.60:62752 | DNS | 52.4 | Isolation Forest |
| 26 | 2032-12-08 12:17:12 | 192.168.33.10:80 | 192.168.33.60:59369 | TCP | 97.3 | Isolation Forest |

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|---|---|---|---|---|---|---|
| 27 | 2032-12-08 12:18:08 | 192.168.33.1:53 | 192.168.33.20:55835 | DNS | 53.0 | Isolation Forest |
| 28 | 2032-12-08 12:16:43 | 192.168.33.10:80 | 192.168.33.50:51043 | TCP | 95.3 | Isolation Forest |
| 29 | 2032-12-08 12:18:30 | 192.168.33.1:53 | 192.168.33.60:53057 | DNS | 51.3 | Isolation Forest |
| 30 | 2032-12-08 12:18:29 | 192.168.33.1:53 | 192.168.33.60:64282 | DNS | 52.4 | Isolation Forest |

# 3. Anomaly Analysis (Isolation Forest)

## Model Configuration

| Parameter | Value |
|---|---|
| Algorithm | Isolation Forest (scikit-learn) |
| Estimators | 100 |
| Contamination | 0.05 |
| Features Used | 20 |

## Top Anomalous Flows

| Source IP | Dest IP | Port | Proto | Packets | Bytes | Score |
|---|---|---|---|---|---|---|
| 192.168.33.10 | 192.168.33.30 | 59637 | TCP | 2 | 334 | **93.4** |
| 192.168.33.1 | 192.168.33.30 | 61066 | DNS | 1 | 92 | **63.3** |
| 192.168.33.10 | 192.168.33.20 | 63745 | TCP | 2 | 342 | **90.5** |
| 192.168.33.10 | 192.168.33.30 | 50306 | TCP | 1 | 176 | **53.7** |
| 192.168.33.1 | 192.168.33.40 | 57788 | DNS | 2 | 172 | **85.1** |
| 192.168.33.1 | 192.168.33.20 | 51067 | DNS | 1 | 92 | **53.0** |
| 192.168.33.1 | 192.168.33.60 | 59557 | DNS | 2 | 157 | **77.6** |
| 192.168.33.1 | 192.168.33.30 | 59114 | DNS | 1 | 92 | **63.3** |
| 192.168.33.10 | 192.168.33.40 | 51358 | TCP | 2 | 344 | **84.9** |
| 192.168.33.1 | 192.168.33.30 | 49483 | DNS | 1 | 92 | **63.3** |
| 192.168.33.10 | 192.168.33.50 | 50527 | TCP | 2 | 344 | **90.8** |
| 192.168.33.1 | 192.168.33.60 | 59763 | DNS | 1 | 72 | **51.3** |
| 192.168.33.1 | 192.168.33.30 | 51273 | DNS | 1 | 72 | **60.3** |
| 192.168.33.10 | 192.168.33.60 | 57544 | TCP | 2 | 341 | **89.3** |
| 192.168.33.10 | 192.168.33.30 | 56238 | TCP | 1 | 176 | **53.7** |
| 192.168.33.10 | 192.168.33.40 | 52606 | TCP | 2 | 331 | **91.9** |
| 192.168.33.1 | 192.168.33.30 | 57026 | DNS | 1 | 72 | **60.3** |
| 192.168.33.10 | 192.168.33.20 | 61314 | TCP | 2 | 335 | **89.6** |
| 192.168.33.1 | 192.168.33.20 | 60827 | DNS | 1 | 92 | **53.0** |

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|---|---|---|---|---|---|---|
| 192.168.33.10 | 192.168.33.30 | 49978 | TCP | 1 | 176 | 53.7 |

## 4. JA3 TLS Fingerprint Analysis

JA3 hashes the TLS ClientHello to identify malware without decryption.

*No TLS ClientHello packets detected in the capture.*

## 5. IP Reputation Analysis

*No external IPs to check.*

## 6. Protocol Distribution

## 7. Geographic Analysis

## 8. Statistical Baseline Deviation

NetSentinel builds a personalized baseline from your traffic to reduce false positives.

## 9. Recommendations

Investigate the top anomalous flows for potential data exfiltration, C2 communication, or lateral movement.

Cross-reference anomalous IPs with your organization's threat intelligence feeds.

Continue monitoring network traffic to refine the behavioral baseline and reduce false positives over time.

Ensure all endpoints have up-to-date security agents and EDR telemetry enabled.

Review firewall and IDS/IPS rules to incorporate indicators from this analysis.

# 10. Chain of Custody

Cryptographic hashes ensure evidence integrity for legal/academic use.

### Evidence Hash (PCAP)

**Algorithm:** SHA-256
**File:** knoten_in_the_wire.pcap
**SHA-256:** N/A (Live capture — no file to hash)
**Computed At:** 2026-07-16T08:06:57.769603+00:00

### Report Hash (PDF)

**Algorithm:** SHA-256
**SHA-256:** Computed after generation
**Generated At:** 2026-07-16T08:06:57.769595+00:00

# 11. Analyst Notes

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Generated by **NetSentinel v1.0** — Encrypted Network Traffic Anomaly Detector & Forensic Analyzer
This report was generated automatically. All findings should be verified by a qualified analyst.
Report integrity can be verified using the SHA-256 hashes in the Chain of Custody section.